

Institute of Actuaries of India

Subject SA3 – General Insurance

September 2016 Examination

INDICATIVE SOLUTION

Solution 1:**i) Reasons to use an internal capital model:**

- The prescribed approach may not well represent the risk profile of a firm.
- An internal model may lower regulatory capital requirement.
- A firm may gain a better understanding of risks faced by modelling them in detail.
- A firm may be able to use capital more efficiently if the capital more closely reflects the risks faced.
- Using an internal model may give a firm credibility in the market and with rating agencies.

Considerations for internal model:

- What are the costs of developing the internal capital model to a standard that the regulator accepts (and cost to maintain it)?
- How will a capital model integrate with existing systems?
- Should we develop a model from scratch or purchase vendor software?
- Do we have the appropriate resources to do this ourselves, or do we need to recruit people / use consultants?
- What if the results of the internal model capital calculation result in a higher capital amount than the prescribed approach?
- Will this give us a competitive advantage in the market to our peers? Alternatively, will we be left behind by not using this type of model?
- Do we want to go through the extensive regulatory review process?

[5]**ii) Underwriters:**

- What are the business plans for each line of business?
- Is the market softening / hardening?
- How is Global Co positioned in the market?
- How will the underlying claims trend and legislative environment impact future underwriting performance?
- How confident are you of achieving the business plans, how much uncertainty is there around premium volume, loss ratio, rate change?

Reinsurance team:

- What is Global Co's current reinsurance program and are there any gaps in the current reinsurance program?
- Is the reinsurance market expected to be hardening or softening?
- How does Global Co manage its concentration risk?
- Reinsurance - do you have any existing modelling used in purchasing decisions, is there any significant facultative or fronting, are any gaps in cover related to mismatches in wordings or structure

Investment team

- What is Global Co's investment strategy?
- Does Global Co focus on Asset Liability matching?
- What risks are inherent in the current investment mix? (interest rate? equity? Exchange rate? credit risk?)

Risk Management:

- What are the major risks identified and our risk tolerance towards these?
- What internal loss data do we have, details of any 'near misses' etc?
- What external data can we use to describe the business environment and external control factors?
- What scenario analyses do you have in relation to operational risk losses, is there any existing modelling around risk (operational or other)

[6]**iii) Using 2016 financial information may have the following limitations:**

- Capital model should be forward looking. Adopting last year's financial performance does not allow for future business plans (i.e. business growth, rate changes, asset mix etc)
- Actual performance may also be distorted by rare catastrophic events or large claims / large movements in investment returns. Need to give consideration to performance (both insurance and investment returns) over a longer time horizon.
- Financial performance is not the best performance indicator of long tail product. It would be better to analyse performance by underwriting or accident year.

Using this year's budget is more forward looking. However the limitations are

- Capital model should be built on a longer time horizon than 1 year.
- Budget may not reflect central estimates, i.e. they may be stretch / easy targets for management.

[3]**iv) Asset**

- Investment strategy now needs to be revised to take into account much smaller asset base (and capital base)
- May need to restate past performances
- Process for monitoring investment managers will need to be re-evaluated
- Review the quality of the current asset portfolio

Capital Management

- Need to discuss solvency position with IRDA immediately
- Likely to have breached the minimum solvency levels set by the company. May require immediate actions to increase the capital level
- Will need to re-evaluate projections of future MCR/solvency positions, and may need to restate past MCR/solvency positions

- Reconsider return on capital target

Pricing & premium adequacy

- Past premiums will need to be reassessed in light of the potential distortion of investment returns
- Future pricing will need to be reconsidered to help restore the solvency ratio of Global Co
- Future premiums to be calculated to take into account of revised ROE targets and investment returns on technical and shareholder funds.

Risk Management

- Controls over performance monitoring of investments will need to be re-evaluated to understand what may have gone wrong
- Level of risk tolerance may need to be reduced as a result of much lower solvency coverage
- Global Co should consider what revisions to the Risk Management Strategy (RMS) (and implementation of the RMS) will be necessary to satisfy the regulator / shareholders.

[8]

v) One possible approach:

Based on current IRDA insurance and class of business definition

Stage 1 – for each line of business i.e. Fire, Marine, Miscellaneous and Health Insurance, determine the gross and net premiums, gross and net incurred claims.

Stage 2 – Carry out scenario testing using varying percentages of the gross premium for calculation of RSM1 and gross incurred claims for calculation of RSM2.

Stage 3 – Stress test the results using different values of factor A and factor B for each line of business.

Calculate the implied capital requirements and solvency ratio using the revised parameters derived from above.

Comparison can then be made between current solvency margin and expected future solvency margin..

Stage 4 – recommend any changes in factors required. Note that the current prescribed method assumes the same factors will apply to all insurers, thus it is not possible to have a single formula to fit all implied values. Small insurers are likely to have a minimum solvency margin lower than their implied figure and vice versa. A trial and error or optimisation approach needs to be adopted to ensure among all insurers the difference between the prescribed and implied figures are minimised.

Data/information you would require:

- Historic IRDA returns (form KG, HG etc), Appointed Actuary's report and FCR for all or a representative sample of small, medium and large insurers
- Calculated economic capital for each insurer as a way of cross checking the implied capital requirements against the existing economic capital held.

[5]

[27 Marks]

Solution 2:

i) Pricing

- Product will need re-pricing.

Policy Wording and Administration

- Alignment of product wordings may be required.
- Ongoing monitoring of pricing will be required, processes will need to be set up.
- Conversion of existing data mappings to suit the new algorithm will be required.
- There will need to be review and testing of the final rates that are being presented to the market.
- If pricing is to be aligned, this may require changes to front end/back end systems, and changes to the rating engines used.

Distribution and Broker Relationships

- If product design is changing and/or prices are changing - distribution will need to be informed and updated.
- Brokers will need to have changes explained to them.
- New rates will impact sales.

Underwriting

- New guidelines, authorities etc. need to be aligned/rolled out.

Claims

- There may be more info that is required to be collected than is currently the case - used for future pricing/monitoring.
- Claims processes may also be aligned - this may impact future costs of claims.
- Historical claims data will have a discontinuity which will impact reserving process.

Reinsurance

- Reinsurance arrangements may need to be aligned - this may impact costs.
- Need to monitor aggregation.

- Consider potential for reducing reinsurance due to diversification arising from combining two companies.

Monitoring

- There is likely to be requirements to map data to common rating factors – e.g. construction type. Analysts will be required to do this.

Finance

- There may be changes to the mapping of products to the ledger.
- There may be a requirement to review budgets.

Marketing

- Potential re-branding of two products.
- Consideration of impact on existing policyholders and opportunities to market to non-existing policyholders.

[5]

ii) Risk

Pricing will be incorrect.

Mitigant

- Monitoring of renewal and strike rates as well as loss ratios to assess pricing risk by category.
- To do this properly, this should be done at the finest level to see if there are any segments that need changing.

Risk

Implementation will be delayed

Mitigant

- Increased resourcing, careful planning and timetabling, involvement of all levels of functions in timetabling

Risk

Decline in retention rates

Mitigant

- Treat renewal business separately from new business in a manner which encourages renewals
- Actively engage with renewals to enable response to any concerns.

Risk

Not all information will be available for pricing. Existing front end systems may not collect sufficient common rating data.

Mitigant

- Business descriptors can be mapped – some default values may need to be selected.
- Some policies may need to be converted on renewal.

Risk

Loss of key staff

Mitigant

- Adequate resourcing and succession planning.
- Potential confusion in implanting new price and product structure can be ameliorated with good staff communication.

Risk

Strike rates for new business reduce

Mitigant

- Closely monitor, look at current strike rates and examine possible impacts before implementation.
- Engage the distribution channels prior to implementation.
- Understand the current pricing in the market.

Risk

Changes in algorithms cannot be implemented through existing front end systems.

Mitigant

- This should be looked at as part of the project scope.
- IT department should be engaged.

Risk

Brand reputational damage

Mitigant

- Engage with brokers and distribution staff, look at strike rates, understand and communicate benefits of the strategy.
- Seek feedback from policyholders to help gain an understanding of the current market.
- Pricing alignment could impact broker relationships – actively seek to explain and respond to any issues.
- Pricing alignment could invoke conflict among distribution channels – need to check relativities and talk to distribution staff.

Risk

Costs of project exceed expectations.

Mitigant

- Project will be looked after by project team- timelines will be set - these will be monitored.

[5]

iii) JoyCo

Cover	Investment income	Insurance Profit
Fire, General Property and Burglary	0.8%	6.8%
Business Interruption	1.1%	-11.9%
Liability	1.2%	12.2%
Total	0.9%	6.1%

RepCo

Cover	Investment income	Insurance Profit
Fire, General Property and Burglary	1.3%	-15.7%
Business Interruption	1.4%	-19.6%
Liability	0.9%	15.9%
Total	1.3%	-11.6%

Insurance Profit = Underwriting Profit + Investment income where
 Underwriting Profit = Premium – Claims – Expenses

In the table above, investment income (as % of GWP) is calculated as 2% x Net Claim Cost, as a proxy to half a year's investment income. This assumes that all coverages are short tail, and that assets backing business are approximately equal to claims provisions.

Other assumptions for calculating investment income should also be considered e.g. Higher returns for liability which would be expected to have a longer tail and zero for short tail lines.

[4]

iv) JoyCo

Cover	Average premium (Rs)	Total expenses
Fire, General Property and Burglary	1,200	42%
Business Interruption	300	45%
Liability	285.7	29%
Total		40.5%

RepCo

Cover	Average premium (Rs)	Total expenses
Fire, General Property and Burglary	636.4	40%
Business Interruption	200	40%
Liability	300	40%
Total		40%

Profitability

JoyCo

- Total package is above target.
- Made relatively more profitable due to lower loss ratios and lower fixed expenses.
- All segments except for Business Interruption are profitable.
- Cover level profitability is relatively different due to the different allocation of expenses between companies.

RepCo

- Total package is below target.
- Loss ratios higher except for Liability.

- Liability currently very profitable
- Could suggest an intention to subsidise the other covers - especially due to much lower average premium in Fire.

Average Premium

JoyCo

- There are considerable differences between the Fire premiums for each company.
- Could be due to different mix of business, or due to different pricing strategy to subsidise Fire.
- Really need to understand the mix of business to be able to comment on the difference in average premiums.

RepCo

- Fire etc. and BI covers seem to be underpriced, average premiums are insufficient.
- BI unprofitable for both companies – this may be a market issue.
- Liability above target for both companies – if this is because the market is hard, need to be aware of impact of changing market conditions and the underwriting cycle.

Expenses

JoyCo

- JoyCo's total expenses slightly higher than Rep Co's.
- JoyCo seems to be allowing for some sort of marginal costing.
- JoyCo's commission is higher - could be a different distribution channel.

RepCo

- RepCo has used the same expense ratio for all covers, so it may not be properly allocated across covers, which makes comparability and assessment of profitability by cover difficult.

RI costs

Are similar - slightly higher for JoyCo

[3]

v)

Item	Current (Rs mill)	Revised (Rs mill)	% of new GWP
Premium	70	95	
Net claims cost	46.2	46.2	48.61%
RI premium	7.7	10.5	11%
Fixed expenses	7.0	7.0	7.36%
Commission	10.5	14.3	15%
Other variable expenses	10.5	14.3	15%
Investment income	0.9	0.9	0.97%
Actual Profit	-11	3.8	4%

This can be derived as follows:

Current average premium = Rs 70m / 110,000 = Rs 636

Fixed expenses per policy = 10% x Rs 70m / 110,000 = Rs 64

Cost of claims per policy = 66% x Rs 70m / 110,000 = Rs 420

That is, New premium = P such that

$P \times [1 - (11\% + 15\% + 15\%)] - \$64 - \$420 \times (1 - 2\%) = P \times 4\%$,

This gives P = Rs 865

So a 35% expected increase in premium for RepCo to meet its required profit target
[4]

- vi) Increasing the premium significantly could impact business volumes.
Need to assess impact on profitability of selling less business more profitably.
Need to investigate the different premium rates applying for the two companies –
Is the insured profile different? Are different perils covered? Could consider restricting cover by excluding certain perils or introducing sub-limits.
Could evaluate existing insured attitude to a change in coverage or separate options for expanded cover.

[2]

[23 Marks]

Solution 3:

i) Common components

- Case Reserves or Individual outstanding claim reserves: Mostly intimated by the ceding insurer. For proportional these could be a portfolio level whereas excess of loss arrangements, individual case reserves are required
- Additional case reserve: These are reserves set aside at the reinsurer's end over and above the ceding insurer's case reserve. These are particularly used to pad up the excess of loss individual case reserves.
- IBNER: This based on the actuarial analysis of the historical adequacy of case reserves plus additional case reserves to estimate the future development of the reserves on known / reported claims and if additional reserves are required
- Pure IBNR: Actuarial estimate of incurred but unreported yet claims
- ULAE / CME reserves: reserves for claim management expenses either actuarially estimated separately or notionally taken as a percentage of the outstanding claim reserves. E.g, management personnel and other defence/legal costs
- Prudential loading: The loading could be driven by regulatory requirements or targeted levels of adequacy under a set of assumptions or purely for additional prudence

[5]

ii) The significant long tail is due to the reporting and settlement delays that can arise for the reinsurer from the following:

- Client/Policyholder – the hospitals and medical practitioners:
 - Internal policies of the hospital in recognising / intimating claims to insurer
 - Since limitation period does not usually apply unless the claimant is expected to be reasonably aware of it, some claims can come years later when the claimant becomes aware
 - Sometimes the hospital fails to detect a potential claim for e.g., in cases of apologies / mediation the insured may become complacent in intimating to the insurer
 - Some cases where minors are the victims, the period of report can be very long (until they attain the legal age to sue) Parents signing liability waivers may not be enforceable in the courts
 - Other internal practices of the insured, viz.,
 - Communication and transparency
 - Recording practices of case history
- Insurer / Ceding company:
 - General administrative time-lag in reporting claims to reinsurer e.g., half-yearly reporting of bordereau to reinsurer would on an average mean a three months additional delay for the reinsurer
 - Occurrence based policies would be characterised by extreme delays in claims reporting as compared with claims made ones
 - Cases of NBNR claims or notified but not registered claims could be one of the chief reasons for claims not appearing in books of insurer and hence the reinsurer that could arise due to
 - Lack of adequate information / documentation to make an estimate
 - Delays in hiring legal counsel / obtaining legal opinion
 - Claims not asserted though reportable to the insurer, the insurer may not set aside a provision
 - Gathering information could get more difficult in complex cases e.g., external parties are also involved but not parting with information fearing their own liability, or time taken for review and investigation
 - In some cases the insurer sets aside case reserves but may not intimate the reinsurer because the initial gross estimate is lower than the reinsurance deductible, if any, but reserves could get revised upward at a future date which could reflect a reinsurer's share. Such cases could have a particularly pronounced impact on reporting delay to reinsurer in cases of excess of loss arrangements.
 - The expertise of the legal counsel is also one of the key factors in helping speedier settlement of claims

- Courts / judicial procedures
 - Pace of the judicial systems could lead to inordinate delays in settlement
 - Even if a lower court gives a verdict the potentiality of appeals could lead to further uncertainty in final settlement
- Other external factors:
 - Increased general awareness amongst the people leading to claims being intimated much later)
 - Changes and reforms in tort laws could lead to delayed reporting accompanied with a sudden spike in claim reports

[10]

iii) (a)

i] Chain Ladder method:

- Pros: If the deductible is low the reinsurer could have got a relatively more claim information
- More useful method where the judicial system is swift and settlements faster
- If industry data available, benchmarking development factors could help in arriving at a more credible reserves estimate
- Cons: A few large claims or spikes in claims could greatly alter the triangle
- Paid and incurred development factors could be significantly different in these product lines
- Ten years is not adequate time period for a medical mal practice line to be fully run-off – estimating tail factors could be a significant challenge though industry benchmarks may be used.

ii] B-F method

- Pros: Where there is no adequate claims experience, can prove to be a reliable blended exposure based method to reserving
- Normally would be a more conservative estimate in the initial years than chain ladder for lines like medical malpractice characterised by slow reporting and settlement
- Cons: The reinsurer needs to rely on the insurer's advice for ULR and the reinsurer may not have full access to insurer's rating rationale
- Being an exposure based measure, adjusting for inflationary trends could be a challenge
- Could be an inappropriate method if there is enough amount of claims information.
- As rates could harden/soften frequently ULR needs to be correspondingly revised that could lead to lesser consistency in year on year reserving basis – compare primary layer ULR with working layer loss ratio
- Estimating tail factor a challenge here as well

iii] Frequency severity method

- The method is more suitable for mass product lines that are volume driven where average claims cost can be statistically estimated from large volumes of claim data
- Both frequency and severity are difficult to determine in classes like medical malpractice as these are usually characterised by a few claims and unpredictable claim amounts
- The frequency itself is a function of policy terms and deductibles that could vary year on year
- Particularly where liability covers are packaged it would be difficult to segregate the frequency of medical malpractice per unit of exposure
- Changes in definitions of claim / incident could also impact the frequency

[9]

(b) This could indicate the Chief Actuary's approach to have a blended result taking into account loss development factors (experience based) and the Pricing loss ratios (exposure based). Ten years of experience for liability class line like medical malpractice is not sufficient to rely solely on chain ladder or frequency-severity based method.

The choice of higher of the two estimates indicates a prudent call by the Chief Actuary.

[2]

(c) Premium information:

- Analysing Earned Premiums and Exposure...
- ... by mix of business e.g., by geography / jurisdiction; by intermediary; type of policy i.e. occurrence or claims made...
- ...by type of reinsurance treaty – oblig / fac; excess of loss / proportional;
- Analyse target pricing loss ratios of the cedant by year of underwriting...
- ... as compared with trends in average premium rate
- Analyse changes in coverage by regulation or otherwise by underwriting year e.g., capping damages
- Year on year changes in retention / deductible
- Extensions / endorsements and premium break-up for them with target pricing loss ratios
- Analyse treaty extensions and aligning the varied maturities of the underwriting years due to these extensions

Claims information:

- Paid and incurred claim information... numbers and amounts...
- ... by underwriting year, reporting year and accident year...
- ... bifurcated into claims closed with and without payment...

- ...by homogeneous classification of premium / exposure as given above
- Changes in claim philosophy/claim administration by reporting year...
- ... changes in key claim personnel and claim processing/settlement ideology...
- ...and analysing noticeable impact of the above on claim triangles
- Changes in case reserving methods by accident year...
- ...bifurcated into indemnity and claim related expenses...
- ... individual case estimates for excess of loss programmes

[6]

iv] a) Stability Clause

A stability clause, or an indexation clause, allows for inflation-related increases in the costs of claims between the ceding insurer and its reinsurer at a given % of a fixed amount

Further analysis/adjustments:

- Indexation clause and effective date for indexing...
- ... any changes in indexation clause year on year e.g. what is the trigger? Policies effective from the date or claims reported from the date..
- ... and changes in indexation method – e.g. by a fixed percentage or a fixed amount
- de-trend the indexation amounts / factors to estimate reserves by the chosen cohort...
- ... apply the indexation factor on the de-trended reserves..
- ... compare inherent inflation with inflation indexed

b) Aggregate Deductible Clause

It is the annual cumulative deductible the ceding insurer has agreed to retain for losses pertaining to the year, as per the treaty terms & conditions.

Further analysis/adjustments:

- Changes in the amounts over various years of underwriting...
- ... how this amount links with the stability clause for each year...
- ... claim frequency with and without AAD
- Adjustments required for estimating LDFs
- For triangulation may need to use all claims i.e. before AAD otherwise LDFs may not be smooth

c) Commutation Clause

Provision in a reinsurance agreement that allows for payment of cash for releasing from all future obligation to pay claims after a certain time period

Further analysis/adjustments:

- Analysis of past commutations if any...
- ... how date of commutation compares with sunset date...
- ...divide contracts with and without the clause...

- ...terms of commutation whether only for known claims (protect against settlement delay only) or all claims

d) Run-off clause

Cancellations on run-off basis mean the reinsurer continues to be responsible for all liabilities pertaining to policy exposures before the date of cancellation.

Further analysis/adjustments:

- LDFs by accident year could be affected due to the abrupt truncation of policy level data
- .. need to analyse interaction between this clause with the other clauses viz., stability, commutation and sunset for assessing IBNR
- Period itself of the run-off provision and any changes in this period over the underwriting years
- IBNR may not be required for reports after the runoff period.

e) Loss Ratio Corridor

A loss ratio corridor is a provision that forces the ceding company to retain losses that would otherwise be ceded to the proportional reinsurance treaty

Further analysis/adjustments:

- Analyse by applicable cohorts...
- ...100% of the corridor or a specific %..
- ... any changes in the % over the years...
- Though proportional reinsurance, pricing ULR could be quite different..
- ... LDFs with and without corridors and ULR

f) Sunset Clause

A clause in a casualty excess of loss reinsurance cover that provides that the reinsurer will respond only to losses reported before some predetermined future date (sunset). The clause is used to limit the reinsurer's exposure to the "long-tail" of liability exposure

Further analysis/adjustments:

- UWY analysis of date of sunset...
- ... benchmarking industry tail factors with the period to sunset...
- ...divide contracts with and without the clause...
- IBNR would not be required after sunset

[18]

[50 Marks]
